

Christopher A. Jones

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PROFILE

Early-career systems and security engineer graduating Dec 2026. Experienced in Linux infrastructure (Debian), systems tooling (Go, Python), and network troubleshooting. Hands-on with systemd, Docker and technical support.

EDUCATION

West Virginia University | Morgantown, WV | Honors College | Expected December 2026

B.S. Computer Science and B.S. Cybersecurity | GPA: 3.79 | President's List (Fall 2025)

Relevant coursework: Operating System Structure, Computer System Concepts, Secure Software Development, Analysis of Algorithms, File and Data Structures

TECHNICAL SKILLS

Linux & systems: Debian 13, Ubuntu Server, Arch; systemd, SSH, users/groups/permissions, apt, storage/filesystems, Apache HTTP Server

Networking & troubleshooting: TCP/IP, DNS, ARP, ICMP, Wireshark, Linux networking, local firewalls, hardware/software troubleshooting

Automation & programming: Bash, Python, Go, C, Java, PowerShell; Ansible, Git/GitHub

Infrastructure & platforms: Docker, Docker Swarm, QEMU, Apache Spark, Apache Hadoop, MySQL, .NET 8

SELECTED SYSTEMS PROJECTS

Self-Hosted Homelab | Personal infrastructure | 2025-present

Proxmox | Debian | Containers | DNS | Tailscale | Observability

- Run Proxmox VE on Pollux and Debian on Castor; operate Pi-hole, Jellyfin in LXC, and a NeoForge Minecraft server. Pollux is a Tailscale subnet router for internal service access.
- Monitor workloads with Prometheus, Grafana, Loki, and Grafana Alloy; use private home.arpa DNS and the home LAN for service troubleshooting and portmango experiments.

portmango: Network Mapper in Go | Personal | 2026-present

Go | TCP/IP | ARP | ICMP | Wireshark | Docker

- Built a Go network mapper with ARP discovery, TCP SYN scanning, and ICMP probing; validated packet behavior in Wireshark against lab targets.

PICARD: Distributed Machine-Learning Platform | Senior capstone, 5-person team | Spring 2026-present

React | .NET 8 | Apache Spark | Apache Hadoop | Docker Swarm | MySQL | OAuth 2.0

- Developing a platform that authenticates and executes researcher-submitted Scala/Python scripts across a Hadoop/Spark cluster.
- Built a Parameter Sweep Engine for experiment batches and created documentation to simplify distributed stack operations.

MergeConflict (MPX): Operating System from Scratch | CS 450 team project | Fall 2024

C | QEMU

- Coded an x86 OS from scratch (no stdlib) including scheduling, memory allocation, and device I/O, debugging under QEMU.

EXPERIENCE

Tutor, LCSEE Learning & Mentoring Center | West Virginia University | Sept 2024-present

- Diagnose programming and systems issues for CS/Cybersecurity students, explaining technical concepts and coaching independent debugging.

Robotics Instructor Intern | NASA Katherine Johnson IV&V Education Resource Center | Summers 2023 & 2024

- Taught VEX IQ robotics, designed STEM activities, and helped organize a regional drone competition; focused on troubleshooting and communication.